| | | | | | |
|---|---|---|---|---|---|
| **8** | **(a)** | | $hc/\lambda = \Phi + E_{MAX}$ | M1 | |
| | | | $h$ = Planck constant, $c$ = speed of light / e.m. radiation | A1 | [2] |
| | **(b)** | **(i)** | gradient of line is $hc$ | M1 | |
| | | | $h$ and $c$ are both constants | A1 | [2] |
| | | **(ii)** | $\Phi = 2.28 \times 1.6 \times 10^{-19}$ | C1 | |
| | | | $= 3.65 \times 10^{-19}$ (J) | | |
| | | | $hc/\lambda_0 = 3.65 \times 10^{-19}$ | | |
| | | | $\lambda_0 = (6.63 \times 10^{-34} \times 3.0 \times 10^{8})/(3.65 \times 10^{-19})$ | C1 | |
| | | | $= 5.45 \times 10^{-7}$ m | A1 | [3] |